

HALLANDSASEN, Sweden

WMO: 026020

Lat: 56.4040N

Lon: 12.9436E

Elev: 522

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.3	13.8	5.3	7.4	21.8	8.3	8.6	31.3	14.5	34.3	13.2	34.3	1.8	230	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	78.6	64.9	75.4	63.2	72.4	61.6	67.2	74.8	65.5	72.3	63.8	70.0	3.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	93.4	70.3	62.9	87.5	68.5	61.2	82.2	67.2	32.0	74.8	30.6	72.4	29.4	70.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.0	8.9	DB	3.7	83.3	4.4	3.4	0.5	85.7	-2.1	87.7	-4.5	89.6	-7.7	92.1	(4)
(5)			WB	3.3	70.2	4.1	2.2	0.3	71.8	-2.0	73.1	-4.3	74.3	-7.3	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	30.0	31.1	35.4	43.9	51.3	57.0	61.9	61.1	54.2	46.5	39.8	33.3	(6)	
	DBStd	12.84	8.00	6.96	6.97	6.37	6.25	4.98	4.66	4.60	5.01	6.35	6.65	8.32	(7)	
	HDD50	2847	621	530	453	202	55	4	0	0	17	140	308	517	(8)	
	HDD65	7152	1086	950	918	632	426	246	122	138	324	572	756	982	(9)	
	CDD50	1221	0	0	1	20	96	212	368	345	144	33	2	0	(10)	
	CDD65	52	0	0	0	0	2	5	25	19	1	0	0	0	(11)	
	CDH74	407	0	0	0	0	25	73	216	91	2	0	0	0	(12)	
	CDH80	40	0	0	0	0	2	5	29	4	0	0	0	0	(13)	
(14) Wind	WSAvg	3.6	4.4	4.1	3.9	3.7	3.5	3.0	2.7	2.5	2.9	3.7	4.5	4.1	(14)	
(15) Precipitation	PrecAvg	32.2	2.5	2.0	1.8	1.6	2.2	3.2	3.5	3.9	3.0	3.1	2.6	2.7	(15)	
	PrecMax	43.4	6.1	5.0	4.0	3.3	4.8	6.4	9.6	7.7	5.4	6.0	4.9	5.4	(16)	
	PrecMin	22.4	0.1	0.2	0.6	0.2	0.8	0.0	0.1	0.9	0.7	1.2	0.7	0.6	(17)	
	PrecStd	5.1	1.5	1.0	0.9	0.9	1.1	1.7	2.3	1.7	1.3	1.3	1.1	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	46.3	58.6	69.8	77.2	80.1	83.5	79.9	72.3	62.0	54.0	49.5	(19)	
		MCWB	45.0	44.4	48.6	56.3	63.2	65.1	68.1	63.7	58.9	54.7	52.9	48.5	(20)	
	2%	DB	43.8	43.7	51.5	64.0	71.8	75.9	79.5	76.5	67.9	58.5	52.0	47.4	(21)	
		MCWB	42.4	41.8	45.1	52.0	58.6	62.5	66.1	64.3	59.2	53.5	50.8	46.3	(22)	
	5%	DB	42.1	41.9	46.7	59.9	67.4	72.3	76.2	73.1	64.9	56.7	50.0	45.2	(23)	
		MCWB	40.8	40.4	42.3	49.3	56.4	60.4	64.8	62.9	56.6	52.8	48.6	43.9	(24)	
	10%	DB	40.2	40.1	44.0	55.6	62.8	68.1	72.8	69.7	62.2	55.0	48.3	43.5	(25)	
		MCWB	39.0	38.7	41.3	46.8	53.5	57.7	62.7	61.3	55.4	51.9	46.9	42.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	44.6	48.7	57.2	65.2	67.3	70.8	69.7	63.1	57.5	53.6	48.6	(27)	
		MCDB	46.1	46.1	56.5	67.8	75.0	77.5	79.0	75.5	68.3	59.2	53.9	49.1	(28)	
	2%	WB	42.8	42.2	45.3	52.9	61.2	63.9	68.0	66.4	60.7	55.3	50.9	46.5	(29)	
		MCDB	43.6	43.1	49.8	62.2	68.3	72.9	76.5	72.0	66.2	57.4	51.7	47.1	(30)	
	5%	WB	40.9	40.4	43.3	50.4	57.9	61.8	66.3	64.9	58.9	53.8	48.8	44.0	(31)	
		MCDB	41.8	41.6	45.9	57.7	64.9	69.5	73.7	70.4	62.9	56.0	49.8	45.0	(32)	
	10%	WB	39.2	38.7	41.4	47.8	55.0	59.6	64.4	63.4	57.2	52.3	47.1	42.4	(33)	
		MCDB	40.1	40.0	43.7	53.9	61.1	66.3	70.8	68.4	60.7	54.4	48.1	43.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	7.9	12.0	17.3	17.2	16.9	16.3	14.7	13.4	10.9	7.3	7.1	(35)	
		MCDBR	6.7	7.1	16.4	23.7	24.2	24.9	23.9	19.7	16.4	10.9	7.4	6.2	(36)	
	5% WB	MCWBR	6.8	6.0	11.0	13.9	14.1	13.7	12.9	10.4	9.8	7.7	6.7	6.4	(37)	
		MCWBR	6.5	6.7	14.0	21.6	21.1	21.5	20.2	16.8	15.4	10.2	7.0	6.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.325	0.345	0.379	0.377	0.369	0.391	0.391	0.360	0.346	0.325	0.292	(40)	
		taud	2.276	2.313	2.342	2.310	2.376	2.421	2.374	2.379	2.450	2.413	2.331	2.251	(41)	
	Ebn at Noon		188	232	258	266	274	277	267	259	252	222	177	158	(42)	
		Edh at Noon	16	24	30	35	35	34	35	33	27	22	16	13	(43)	
(44) All-Sky Solar Radiation	RadAvg		131	317	748	1259	1644	1727	1602	1289	906	436	164	89	(44)	
	RadStd		25	51	77	168	158	148	192	110	103	66	30	10	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-292	N/A	N/A

Nomenclature: See separate page